DSA 210 Project Proposal

Project Title

The Effect of Clickbait Headlines on Reddit Engagement: An Analysis with News Source Credibility

Motivation & Research Question

This project aims to investigate whether clickbait-style headlines in Reddit news posts lead to higher user engagement, measured by upvotes and number of comments. Additionally, the study examines how the credibility of the news source influences this relationship.

The main research question is: *Do clickbait headlines on Reddit receive higher engagement compared to non-clickbait headlines, and does the reliability of the news source affect this relationship?*

Data Sources

The primary dataset for this study will be the World News Reddit Data dataset obtained from Kaggle. This dataset contains approximately 110,000 Reddit posts from the r/worldnews subreddit, including variables such as title, score (upvotes), upvote ratio, number of comments, timestamp, and URL. The dataset will be directly downloaded from Kaggle and used as the main source for analyzing engagement patterns.

To identify clickbait headlines, a second dataset, the Clickbait Dataset from Kaggle, will be used. This dataset includes approximately 32,000 labeled headlines, classified as clickbait or non-clickbait. A classification model will be trained using this dataset and then applied to the Reddit post titles to generate a clickbait label for each post.

Data Enrichment

As part of data enrichment, news source credibility will be incorporated into the analysis. Domains will be extracted from the URLs in the Reddit dataset and matched with a domain list compiled from Media Bias/Fact Check. This will allow each post to be assigned a reliability label (reliable or non-reliable). Additionally, posts that link directly to Reddit will be filtered out to focus only on external news sources.

Planned Analysis

The analysis will begin with exploratory data analysis (EDA) on both datasets. After enriching the Reddit dataset with clickbait and reliability labels, statistical analysis will be conducted to examine the relationship between clickbait headlines and engagement. Furthermore, the role of source credibility in modifying this relationship will be investigated.